

Aravinda Raman Jatavallabha

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EXPERIENCE

AI Engineer | Long Health, Cary, NC (Remote)

Jun 2025-Present

- Architected and deployed **serverless ML pipelines** (AWS Lambda, ECS, S3) powering end-to-end clinical documentation workflows; processed **5K+** healthcare documents weekly and raising throughput **35%** over the prior batch design.
- Designed asynchronous, event-driven inference workflows using RabbitMQ for OCR (AWS Textract), RAG processing, LLM summarization, and ICD-10 inference; reduced processing latency by **30%** with **99.9%** uptime.
- Engineered **RAG-based** impairment scoring and clinical report generation workflows with traceable guideline grounding; achieved score variance within **5%** of clinician-calculated benchmarks.
- Built **Model Context Protocol (MCP)** servers and tool- calling workflows exposing internal data and calculations services to GPT agents, replacing hand-written glue between the LLM layer and platform services.
- Deployed **OpenAI Whisper** for real-time transcription and automated clinical report generation from physician-patient conversations.
- Designed and implemented **OpenAI and Anthropic Claude-powered pipelines** for structured extraction, and context-aware Q&A; tuned chunking and retrieval to curb hallucination, cutting documentation and triage time by **50%**.
- Built and maintained multi-user physician platform (Angular, NestJS) with **4-role RBAC**, WebSocket-based real-time updates (Soketi), Stripe-based billing workflows, and standards-based impairment calculation engine with auditable reporting.

Data Scientist/ML Engineer (Co-op) | SmartProtect Public Safety Solutions, Wilmington, DE (Remote)

May 2024-Jun 2025

- Developed and A/B tested time series forecasting models (**ARIMA, Prophet, LSTM**) on **1.2M+** emergency call records to predict demand surges and optimize staffing, improving scheduling accuracy by **20%**.
- Designed **clustering-based optimization algorithms** for dynamic staff allocation based on call volume trends and anomalies, reducing overtime by **18%** and increasing resource utilization by **22%**.
- Productionized ML pipelines using Flask APIs, AWS SageMaker, and SQL-driven feature engineering with CI/CD automation, reducing retraining time by **35%** and improving deployment reliability.
- Fine-tuned **Azure OpenAI LLMs** and implemented **RAG** over dispatcher transcripts for real-time anomaly summarization and context-aware Q&A, reducing incident triage time by **35%**.

Machine Learning Researcher | Defence Research and Development Organisation, Bengaluru, India

Jan 2023-Jun 2023

- Engineered **Temporal Graph Neural Network (GNN)** leveraging continuous temporal features to predict evolving user interactions, improving model accuracy by **2%** over baseline benchmarks [\[Paper\]](#).
- Integrated **Incremental BERT (iBERT)** with the Temporal GNN to model semantic drift in real-time text streams, achieving **3.19 perplexity (6% improvement over prior SOTA)** in masked language modeling; results published in Springer ICPR 2024 [\[Paper\]](#).

Data Science Intern | Merkle, Bengaluru, India

May 2022-Jul 2022

- Led 4 interns building XGBoost, LightGBM, and LSTM revenue models on 10M+ retail records for a 10% campaign profitability uplift (Home Depot); built PySpark/Snowflake ETL over 16M+ rows, cutting query latency 40%.

PROJECTS & PUBLICATIONS (SELECTED)

- TrustBench: Benchmarking Trustworthy Large Language Models** [\[Paper\]](#): Evaluated GPT-4.1, GPT-4o, GPT-5, and Claude Sonnet 4.5 across stability, ambiguity handling, repeatability, and cost-effectiveness metrics for deployment readiness.
- Prompting for LLM Security and RAG** [\[Paper\]](#): Developed a practitioner-focused survey of LLM prompting techniques and cybersecurity defenses, covering prompt injection, RAG attacks, optimization, and secure prompting strategies.
- Privacy Awareness in Large Language Models: Input Regurgitation and Prompt-Induced Sanitization** [\[Paper\]](#): Studied regurgitation risk and prompt-based sanitization strategies for privacy-sensitive LLM deployments under HIPAA/GDPR constraints.

TECHNICAL SKILLS

- Languages & Backend**: Python, SQL, TypeScript, JavaScript; Flask, NestJS, Angular, React, REST APIs, event-driven services, WebSockets, system design
- LLMs & Agents**: OpenAI API, Anthropic Claude, Model Context Protocol (MCP), LangChain, RAG pipelines, prompt engineering, LLM evaluation & guardrails, prompt-injection defense, hallucination mitigation, vector DBs (ChromaDB, FAISS, Pinecone), Hugging Face, LLaMA, Whisper
- ML & Deep Learning**: PyTorch, TensorFlow, scikit-learn, Transformers, PyTorch Geometric; CNNs, GNNs, LSTMs, time-series forecasting, anomaly detection
- Data & MLOps**: Pandas, NumPy, Airflow, Spark, RabbitMQ, Docker, CI/CD, drift detection, automated retraining, A/B testing
- Cloud**: AWS (Lambda, ECS, S3, SageMaker, Textract, ECR), Azure, Linux, Git; HIPAA and PHI/PII-safe system design, Document AI & OCR, Power BI

EDUCATION

Master of Computer Science, Data Science Specialization | NC State University, Raleigh, NC | GPA 4.0/4.0

May 2025

B. Tech in Information Technology, Big Data Analytics minor | Manipal Institute of Technology, India | GPA 4.0/4.0

Jul 2023